

1. Answer the following short questions in one line (MAX).

- Which traversal method visits the root node first? [1]
Ans.
- Can two different binary trees have the same preorder and postorder traversal? [1]
Ans.
- What is the formula of finding the right child if we are using 1 based indexing? [1]
Ans.
- A skewed binary tree is a tree where: [1]
A) All nodes have two children
B) Nodes are uniformly distributed
C) Each node has only one child
D) Tree is perfectly balanced
- In a binary tree, nodes that share the same parent are called: [1]
A) Ancestors B) Descendants C) Siblings D) Leaves
- What is a binary tree called where every node has either 0 or 2 children? [1]
A) Perfect B) Complete C) Full D) Unbalanced

2. A chocolate factory organizes its production process as a binary tree. Each node represents a batch of chocolates and the value stored in the node indicates the quality score of that batch. A production path is considered valid if the sum of quality scores from the root to that node is divisible by 3. Write a function to determine the level number that contains the highest number of valid batches. If multiple levels have the same maximum count, return the smallest level number. But, if no node satisfies the condition, return -1. Assume the height of the binary tree will not exceed 10 and your level starts from 0.

Constraints: Do NOT use instance variables or static variables.

Sample Input	Sample Output	Explanation
<pre> 3 /\ 1 2 /\ \ 3 0 4 </pre>	0	Level 0: Root (3) → sum = 3 (divisible by 3) → Level 0, so this is a candidate Level 0 has 1 valid path Level 1: Node (1) → 3 + 1 = 4 → not divisible by 3 Node (2) → 3 + 2 = 5 → not divisible by 3 Level 1 has 0 valid paths. Other levels also can have valid paths which are divisible by 3, but as level 0 has a valid path , level 0 is returned according to the constraint.
<pre> 1 /\ 2 4 / \ 5 7 </pre>	1	1 → No → Level 0 1+2=3 (valid path) → Yes → Level 1 1+4=5 → No → Level 1 1+2+5=8 → No → Level 2 1+4+7=12 (valid path) → Yes → Level 2 Candidates are Level 1 and Level 2 so we choose level 1 according to our constraints.

JAVA	PYTHON
<pre> public static int maxValidLevel(TreeNode root){ //ToDo} </pre>	<pre> def max_valid_level(root): #ToDo </pre>